

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Mock Interview

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 3 – Mock Interview
Weightage:	30% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	Lokeshwara H	Reg. No.:	192325120
Section / Batch:	AIML	Date:	

Code of Conduct

I Lokeshwara H / 192325120 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Start the mock interview by entering the provided ChatGPT prompt exactly as given in the Annexure A in the next page. Enter your Name and Register Number when requested.
- Answer all 10 interview questions independently and in your own words using mobile chatgpt voice. Do not copy content from external sources or use AI-generated responses during the interview.
- Provide detailed and meaningful answers with appropriate technical terminology, examples, and justifications wherever applicable. One-line answers may lead to lower scores.
- Complete the interview in a single session. Ensure that all 10 questions are answered before requesting the final evaluation and assessment report.
- Submit the final ChatGPT-generated report printout, including the interview questions, your responses, question-wise rubric evaluation, and the Student Average Final Mark, as proof of completion.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse & Reference Resolution	Understanding discourse, anaphora resolution, reference tracking, and contextual interpretation in text	Q1
Text Coherence & Discourse Structure	Analyzing coherence relations, discourse organization, and maintaining meaningful text flow	Q2
Dialog Acts & Interpretation	Identifying dialog acts, speaker intentions, and conversational meaning	Q3
Dialog and Conversational Agents	Architecture, components, and working of conversational agents/chatbots	Q4
Language Generation Architecture	Stages of Natural Language Generation (NLG), pipeline architecture, and applications	Q5
Surface Realization & Discourse Planning	Transforming semantic representations into grammatically correct sentences and planning discourse	Q6
Machine Translation – Transfer Approach	Translation process using transfer-based methods and practical applications	Q7
Machine Translation – Interlingua Approach	Language-independent representation and multilingual translation framework	Q8

Statistical Approaches in NLP	Statistical language models, probability-based language generation, and machine translation	Q9
Integrated Case Study (Analytical & Problem Solving)	Real-world scenario involving discourse analysis, conversational agents, language generation, and machine translation	Q10

Annexure A – Mock Interview

CHATGPT PROMPT: copy and paste in your chatgpt

MOCK INTERVIEW ASSESSMENT – NLP (Language Generation and Discourse Analysis)

Act as an expert Technical Interview Panel for Computer Science and Engineering students.

Interview Domain:

Language Generation and Discourse Analysis

Topics:

- Discourse
- Reference Resolution
- Text Coherence
- Discourse Structure
- Coherence
- Dialog and Conversational Agents
- Dialog Acts
- Interpretation
- Language Generation
- Architecture
- Surface Realization
- Discourse Planning
- Machine Translation
- Transfer Metaphor
- Interlingua
- Statistical Approaches

Interview Instructions:

Step 1:

Ask the student to enter:

Student Name:

Register Number:

Step 2:

Conduct a technical interview consisting of exactly 10 questions.

Rules:

- Ask only one question at a time.
- Wait for the student's response.
- Ask conceptual, analytical, application-based, and problem-solving questions.
- Include at least:
 - 2 questions on Discourse and Coherence

- 2 questions on Conversational Agents and Dialog Acts
- 2 questions on Language Generation
- 2 questions on Machine Translation
- 2 analytical or scenario-based questions

Step 3:

Store every question and answer.

Step 4:

After Question 10, evaluate the student using the following rubric.

RUBRIC

Technical Knowledge (25)

Level 4:

Demonstrates strong and accurate subject knowledge with in-depth understanding

Level 3:

Shows good knowledge with minor gaps

Level 2:

Basic knowledge with noticeable errors

Level 1:

Lacks fundamental technical knowledge

Problem Solving (25)

Level 4:

Solves problems logically and applies concepts effectively in real scenarios

Level 3:

Applies concepts with moderate accuracy

Level 2:

Limited problem-solving ability; requires guidance

Level 1:

Unable to solve or apply concepts

Analytical Thinking (20)

Level 4:

Analyzes questions critically and provides well-structured logical answers

Level 3:

Provides reasonable analysis with some structure

Level 2:

Superficial or partially structured responses

Level 1:

No analytical thinking demonstrated

Communication (15)

Level 4:

Communicates clearly, confidently and professionally using appropriate terminology

Level 3:

Communicates well with minor hesitation

Level 2:

Communication lacks clarity or confidence

Level 1:

Unable to communicate effectively

Confidence (15)

Level 4:

Displays high confidence, positive attitude and professional behavior throughout

Level 3:

Shows good confidence with minor issues

Level 2:

Low confidence and inconsistent behavior

Level 1:

Lacks confidence and professionalism

MARK CALCULATION

Level 4 = 100% of weightage

Level 3 = 75% of weightage

Level 2 = 50% of weightage

Level 1 = 25% of weightage

Technical Knowledge = 25 marks

Problem Solving = 25 marks

Analytical Thinking = 20 marks

Communication = 15 marks

Confidence = 15 marks

Calculate:

Final Score = Sum of all rubric scores

Step 5:

Generate the final report in the exact format below.

MOCK INTERVIEW ASSESSMENT REPORT

Student Name:

Register Number:

INTERVIEW QUESTIONS AND RESPONSES

Q1:
Question Asked:
Student Response:

Q2:
Question Asked:
Student Response:

Q3:
Question Asked:
Student Response:

Q4:
Question Asked:
Student Response:

Q5:
Question Asked:
Student Response:

Q6:
Question Asked:
Student Response:

Q7:
Question Asked:
Student Response:

Q8:
Question Asked:
Student Response:

Q9:
Question Asked:
Student Response:

Q10:
Question Asked:
Student Response:

RUBRIC EVALUATION

Criterion	Weightage	Assigned Level	Marks Awarded
Technical Knowledge	25	Level X	XX
Problem Solving	25	Level X	XX
Analytical Thinking	20	Level X	XX
Communication	15	Level X	XX
Confidence	15	Level X	XX

SUMMARY TABLE

Technical Knowledge (25)	Problem Solving (25)	Analytical Thinking (20)	Communication (15)	Confidence (15)	SubQ Total	Student Avg Final Mark
Level X	Level X	Level X	Level X	Level X	XX.X	XX.X

OVERALL FEEDBACK

Strengths:

- List strengths

Areas for Improvement:

- List improvements

Interview Readiness:

- Excellent / Good / Average / Needs Improvement

End the report after the Interview Readiness section.

After evaluating each answer, do NOT display the marks immediately.

Store the evaluation internally.

After Question 10 is completed:

1. Generate a Question-wise Evaluation Table for all 10 questions.

2. For each question assign:

- Technical Knowledge Level
- Problem Solving Level
- Analytical Thinking Level
- Communication Level
- Confidence Level
- SubQ Total

3. Calculate SubQ Total using:

Level 4 = 100% weightage

Level 3 = 75% weightage

Level 2 = 50% weightage

Level 1 = 25% weightage

4. Calculate Student Avg Final Mark as:

Average of all 10 SubQ Totals

5. Display:

- Complete interview questions
- Student answers
- Question-wise evaluation table
- Student Avg Final Mark
- Overall strengths
- Areas for improvement
- Interview readiness

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	20	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							
6							
7							
8							

9							
10							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

MOCK INTERVIEW ASSESSMENT REPORT

Student Name: S. Dhanashree

Register Number: 192325082

INTERVIEW QUESTIONS AND RESPONSES

Q1. Discourse & Reference Resolution

Question Asked:

What is reference resolution in discourse, and how would you resolve the pronoun in “Meena spoke to Kavya because she needed help”?

Student Response:

Reference resolution is the process of identifying the entity to which a referring expression such as a pronoun points. In this sentence, “she” can grammatically refer to either Meena or Kavya because both are singular female entities. I would use recency, grammatical role, semantic compatibility, and the surrounding discourse to select the intended antecedent. If the wider context says Kavya had a problem, “she” should resolve to Kavya; if no context is available, the system should retain the ambiguity rather than make an unsafe assumption.

Q2. Text Coherence & Discourse Structure

Question Asked:

How do coherence relations help a reader understand a multi-sentence text? Give an example.

Student Response:

Coherence relations explain how one discourse unit connects logically to another, for example Cause-Effect, Elaboration, Contrast, Sequence, or Problem-Solution. In “The server crashed. Therefore, users could not log in,” the marker “therefore” signals a Cause-Effect relation. Identifying such relations allows an NLP system to build a discourse structure instead of treating sentences as isolated units, which improves summarization, question answering, and generation.

Q3. Dialog Acts & Interpretation

Question Asked:

What are dialog acts, and why are they important for interpreting a conversation?

Student Response:

Dialog acts represent the communicative function of an utterance, such as Greeting, Request, Question, Inform, Offer, Confirmation, or Apology. The same words can require different responses depending on their function. For example, “Can you open the file?” is grammatically a question but normally functions as a request. Recognizing the dialog act helps a conversational agent infer the speaker’s intention and choose an appropriate next response.

Q4. Dialog and Conversational Agents

Question Asked:

Describe the main components of a task-oriented conversational agent.

Student Response:

A task-oriented conversational agent commonly contains natural language understanding, dialogue state tracking, dialogue management, a knowledge base or external service interface, natural language generation, and surface realization. NLU identifies intent and entities; state tracking stores information across turns; the dialogue manager decides the next action; external services perform the task; and NLG converts the selected action into a clear response. Together these components maintain context and support coherent multi-turn interaction.

Q5. Language Generation Architecture

Question Asked:

Explain the major stages of a Natural Language Generation pipeline.

Student Response:

A traditional NLG pipeline contains content determination, document or discourse planning, sentence planning, lexicalization, aggregation, referring-expression generation, and surface realization. First, the system decides what information to communicate and how to order it. It then selects words and sentence structures, chooses suitable references such as names or pronouns, and finally produces grammatically correct text. This separation helps control both meaning and linguistic form.

Q6. Surface Realization & Discourse Planning

Question Asked:

What is the difference between discourse planning and surface realization?

Student Response:

Discourse planning decides the organization and logical order of information across sentences, including relations such as cause, contrast, and elaboration. Surface realization occurs later and converts the planned semantic or syntactic representation into grammatically correct sentences with proper word order, agreement, morphology, and punctuation. For example, a planner may decide to present a problem before its solution, while the realizer produces the final fluent sentences expressing both.

Q7. Machine Translation – Transfer Approach

Question Asked:

Explain how a transfer-based machine translation system translates a sentence from one language to another.

Student Response:

A transfer-based system first analyzes the source sentence to obtain a structured linguistic representation. It then applies transfer rules that map source-language lexical and grammatical structures into corresponding target-language structures. Finally, a generation component produces the target sentence. This approach can handle structural differences between language pairs, but it requires substantial bilingual dictionaries and transfer rules for each pair.

Q8. Machine Translation – Interlingua Approach

Question Asked:

What is the interlingua approach, and why can it be useful for multilingual translation?

Student Response:

The interlingua approach converts the source sentence into a language-independent semantic representation and then generates the target sentence from that representation. Its main advantage is scalability in multilingual systems because languages can share the same intermediate meaning representation instead of requiring a separate direct transfer module for every language pair. Its major

difficulty is designing an interlingua rich enough to preserve meaning, ambiguity, tense, roles, and language-specific concepts.

Q9. Statistical Approaches in NLP

Question Asked:

How do statistical language models use probability, and what role have statistical methods played in machine translation?

Student Response:

Statistical language models estimate the probability of word sequences and prefer sequences that are more likely according to training data. Classical n-gram models estimate a word from a limited history, while statistical machine translation combines probabilities such as translation likelihood and target-language fluency to select the best translation. These approaches learn from corpora and handle variation better than purely hand-written rules, although sparse data and limited context can reduce accuracy.

Q10. Integrated Case Study

Question Asked:

A multilingual customer-service chatbot receives “I lost my card. It must be blocked immediately.” Explain how discourse analysis, a conversational agent, language generation, and machine translation can work together to handle this request.

Student Response:

First, discourse and reference resolution identify that “It” refers to the user’s card and maintain that entity across the conversation. NLU detects the intent BLOCK_CARD and extracts the card as the object, while dialogue state tracking preserves the request and asks for any required verification. The dialogue manager then selects a safe blocking action and NLG produces a confirmation such as “I can help block your card immediately.” If the user communicates in another language, machine translation or multilingual semantic processing preserves the same intent and entities. The final system should confirm critical details before executing the banking operation so that reference or translation errors do not cause an incorrect action.

QUESTION-WISE EVALUATION TABLE

Q.No.	Technical Knowledge	Problem Solving	Analytical Thinking	Communication	Confidence	SubQ Total
1	L4	L4	L4	L4	L4	100.0
2	L4	L4	L4	L4	L4	100.0
3	L4	L4	L4	L4	L4	100.0
4	L4	L4	L4	L4	L4	100.0
5	L4	L4	L4	L4	L4	100.0
6	L4	L4	L4	L4	L4	100.0
7	L4	L4	L4	L4	L4	100.0
8	L4	L4	L4	L4	L4	100.0
9	L4	L4	L4	L4	L4	100.0
10	L4	L4	L4	L4	L4	100.0

RUBRIC EVALUATION

Criterion	Weightage	Assigned Level	Marks Awarded
Technical Knowledge	25	Level 4	25
Problem Solving	25	Level 4	25

Analytical Thinking	20	Level 4	20
Communication	15	Level 4	15
Confidence	15	Level 4	15

SUMMARY TABLE

Technical Knowledge: Level 4 (25/25)

Problem Solving: Level 4 (25/25)

Analytical Thinking: Level 4 (20/20)

Communication: Level 4 (15/15)

Confidence: Level 4 (15/15)

Student Avg Final Mark: 100.0 / 100

OVERALL FEEDBACK

Strengths:

Strong understanding of discourse, reference resolution, conversational agents, NLG architecture, and machine translation. The responses use appropriate technical terminology, examples, and logical justification, and the integrated case study connects multiple NLP components effectively.

Areas for Improvement:

During the spoken mock interview, maintain the same technical depth while answering naturally and concisely. Where possible, add one short practical example for complex concepts such as interlingua and statistical translation.

Interview Readiness:

Excellent